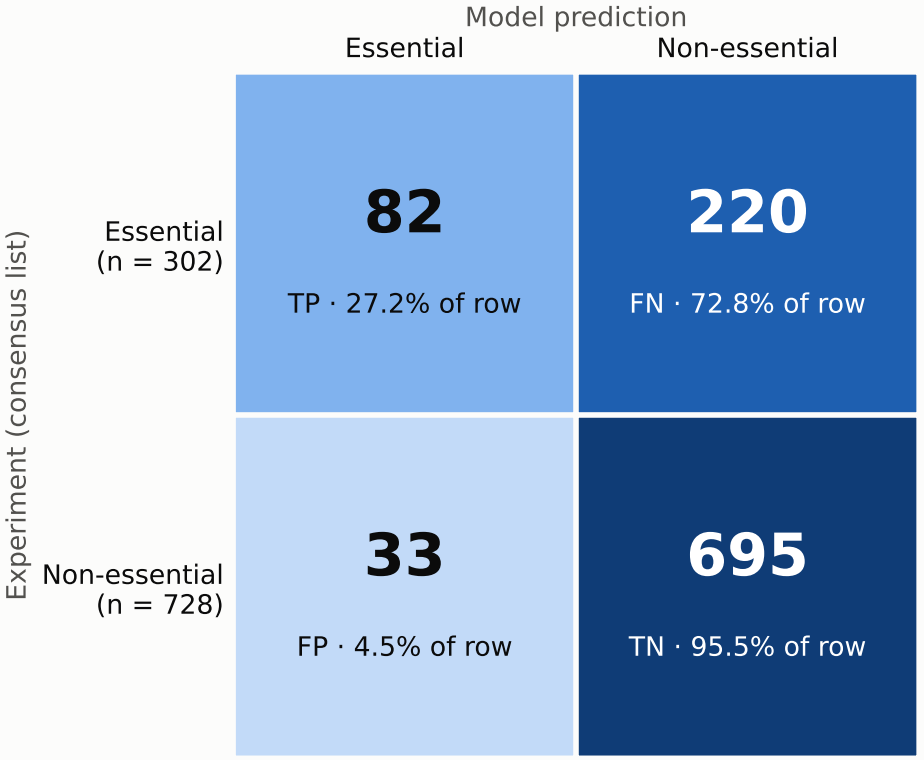
iYali26 newest candidate (a56f3649): single-gene essentiality vs consensus list (KO/WT < 15 %)

Task A medium + PO1f, glucose uptake 2.414 · in-memory diagnostic overlay, assumed caps R1384 ≤ 8 and R860 ≤ 1

Benchmark rule: no-solution KOs unresolved



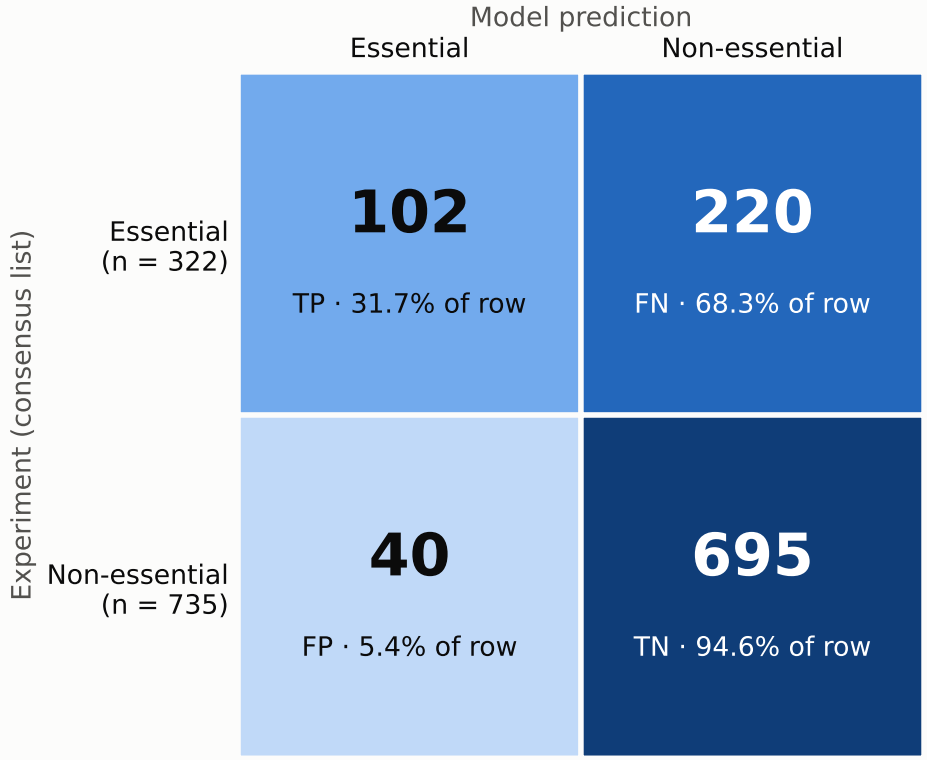
Recall 27.2% · Precision 71.3% · Specificity 95.5% · MCC 0.327

27 no-solution KOs left out · 16 mtDNA excluded

WT 0.3118 h^{-1} · 1,073 genes screened · complex I KO/WT 0.759

Listed positives in model 322 / 1,612 (1,290 out of scope, not FN)

Labelled view: no-solution KOs counted as lethal



Recall 31.7% · Precision 71.8% · Specificity 94.6% · MCC 0.354

27 no-solution KOs counted as lethal; all fail only on maintenance ATP

WT 0.3118 h^{-1} · 1,073 genes screened · complex I KO/WT 0.759

Listed positives in model 322 / 1,612 (1,290 out of scope, not FN)

Static FBA, Task A medium (po1f_csm_leu_w29uptake_20261012.csv) + PO1f strain profile, objective biomass_C; essential = unrounded KO/WT strictly below 0.15.
Overlay = Task A's diagnostic package in memory (not curated); the two caps are assumptions. The benchmark rule (left) is primary; the lethal view (right) is a labelled second view.
Cell shade = share of the experimental row. Negatives are model genes not on the consensus list (a user-defined class, not experimental non-essential calls). The list was used during model development, so this is a development reference, not independent validation. Screen run 2026-10-14.